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THE FLOOR – what does weaker symmetry actually buy?
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A Boolean function's SENSITIVITY is the most single-bit flips that can change its answer, at the worst input. Low sensitivity means stable.

Impose a symmetry and you constrain how low it can go. Two symmetries:

FULL – f depends only on how many 1s the input has (any reordering)
 CYCLIC – f is unchanged by rotating the input (the 3-6-9 shift map)

Cyclic is strictly weaker, so it should permit lower sensitivity.
 The question is whether it actually does, and by how much. Both sides below are exhaustive: every function counted, nothing sampled.

[1] THE TWO FLOORS, BOTH EXACT

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n	full-symmetry floor	cyclic floor	gain	status

3	2	2	0	both exact
4	3	2	1	both exact <-- cyclic WINS
5	3	3	0	both exact
6	4	3	1	both exact <-- cyclic WINS
7	4	3	1	both exact <-- cyclic WINS

full-symmetry floor alone, pushed further (still exact):

n	floor	ceil((n+1)/2)
3	2	2
4	3	3
5	3	3
6	4	4
7	4	4
8	5	5
9	5	5
10	6	6
11	6	6
12	7	7
13	7	7
14	8	8
15	8	8
16	9	9

It tracks ceil((n+1)/2) exactly – full symmetry forces sensitivity

to grow LINEARLY. That is the wall cyclic symmetry has to beat.

[2] THE FUNCTION THAT BEATS SYMMETRY (n = 6)

full symmetry cannot get below 4. Cyclic reaches 3.

36 functions achieve it. Here is one, written out completely:

necklace 000000	weight 0
necklace 000001	weight 1
necklace 000011	weight 2
necklace 000101	weight 2
necklace 000111	weight 3

weights used: [0, 1, 2, 2, 3]

and here is why it is NOT symmetric – weight 2 has three necklaces, and this function accepts some but not all of them:

000011	ACCEPT
000101	ACCEPT
001001	reject

A symmetric function must treat all three the same. This one does not. That is exactly the extra freedom cyclic invariance provides, and it is worth a full unit of sensitivity here. Small, but real and exact.

[3] THE HONEST LIMIT OF THIS EXPERIMENT

n = 7 is the end of exhaustive reach: 2^{20} functions is a second, 2^{36} is a week, 2^{60} is never. And the earlier attempt to push past it with annealing FAILED ITS OWN AUDIT -- rerunning with ten times the compute lowered the reported floor at n = 9 and n = 10, which means those numbers were measuring the search, not the mathematics. They are withdrawn.

So the honest state of this line:

n = 3..7	cyclic floor	2, 2, 3, 3, 3	EXACT
n >= 8	unknown to us;	our search is not trustworthy at that size	
full sym	ceil((n+1)/2)		EXACT and linear

The gap at n = 6 and n = 7 is one. Whether that gap widens -- whether cyclic symmetry eventually buys the dramatic savings the literature reports for large n -- is not visible from where we can compute.

VERDICT

Did we find new mathematics? No.

Did we find something real? Yes, three things, all checkable:

1. Full symmetry forces sensitivity $\geq \text{ceil}((n+1)/2)$. Exact, all n tested, and it means symmetric functions are never stable.
2. Cyclic symmetry genuinely beats that -- by one, at $n = 6$ and 7 -- and we can display the exact function that does it and point at the precise weight-2 necklace it treats asymmetrically.
3. Our own search was unreliable and we caught it by re-running with more compute. That check is the reason the table above contains no numbers we cannot defend.

Point 3 is the one I would keep. An experiment that catches its own bad numbers is worth more than one that reports impressive ones.

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